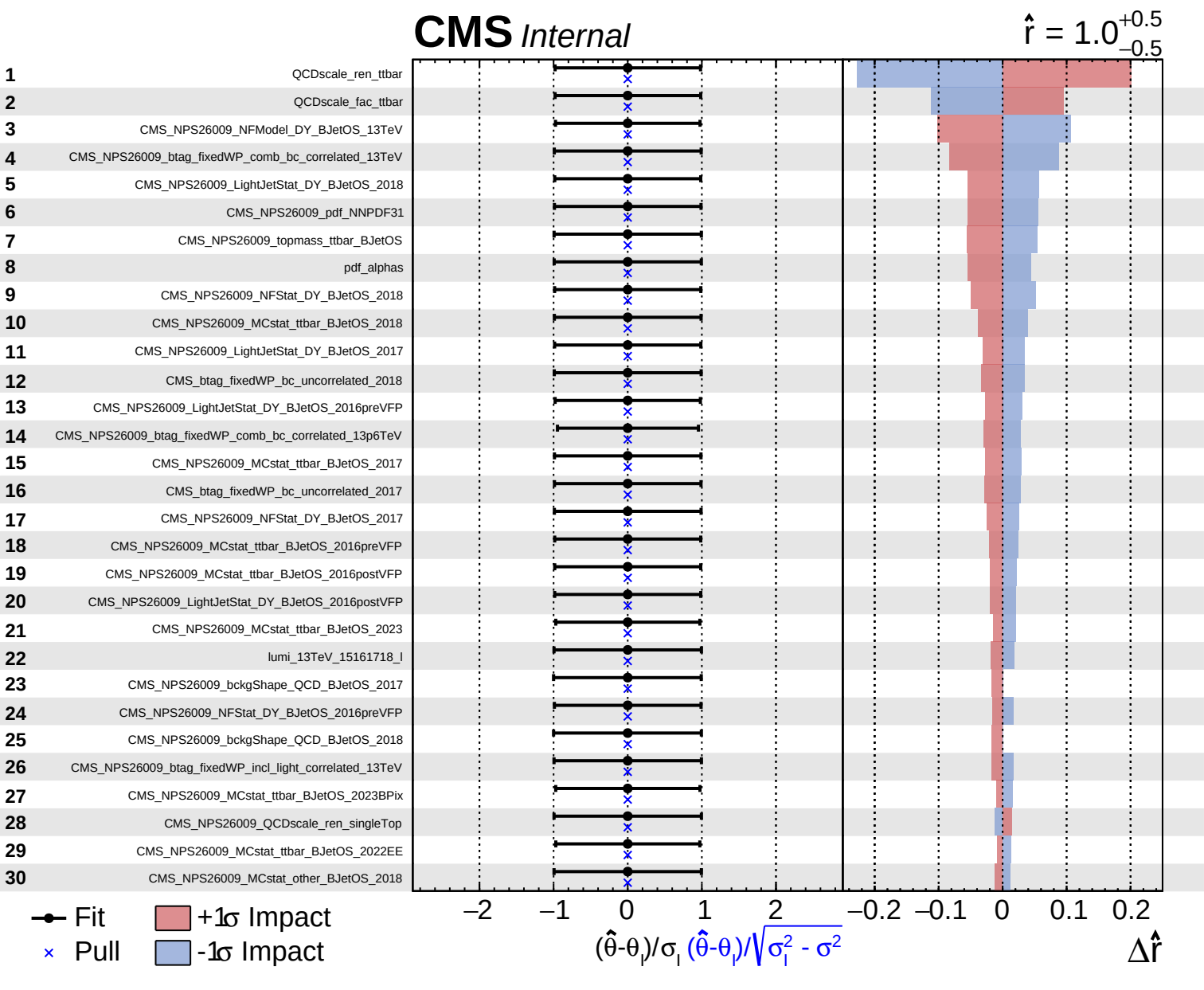
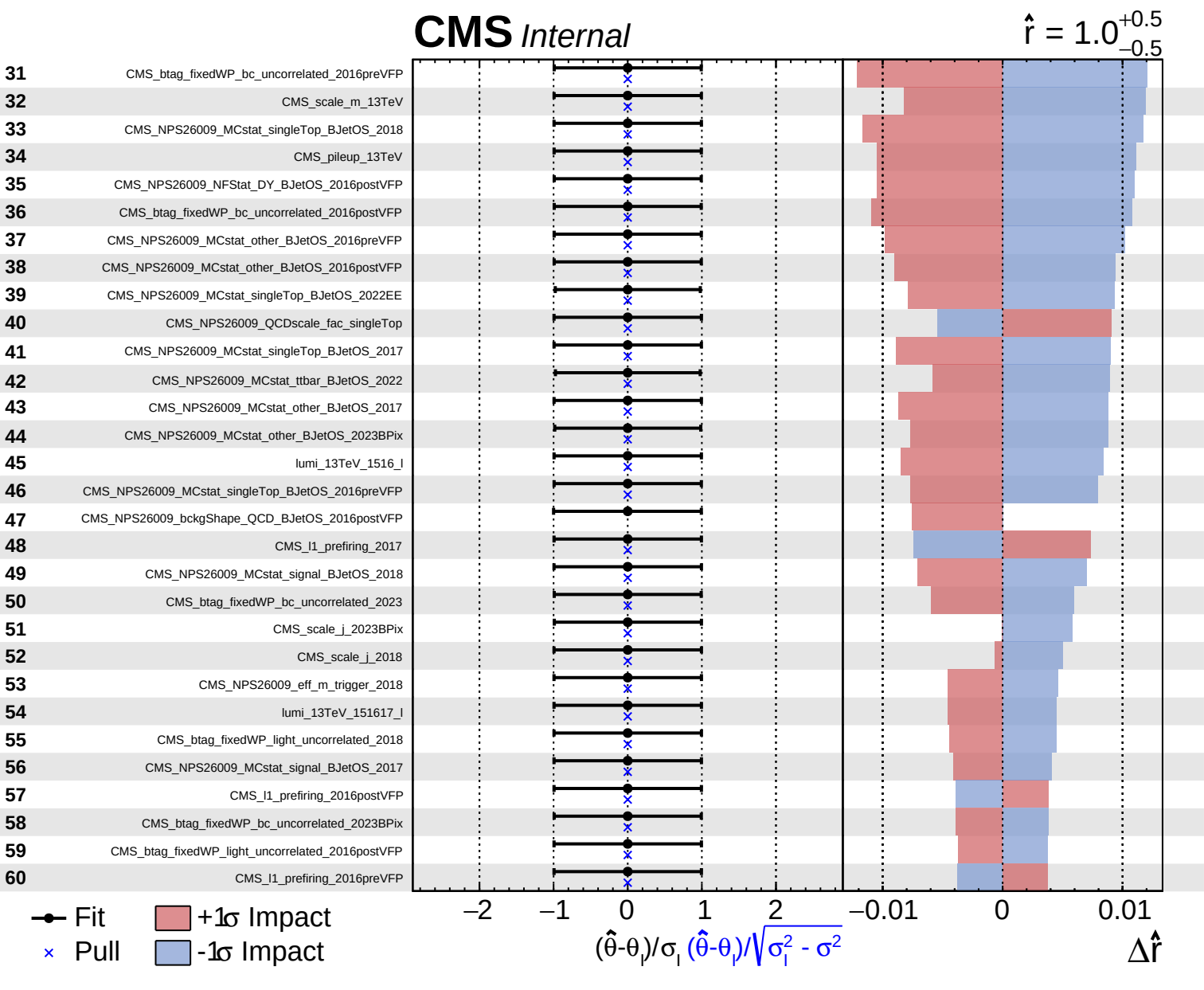
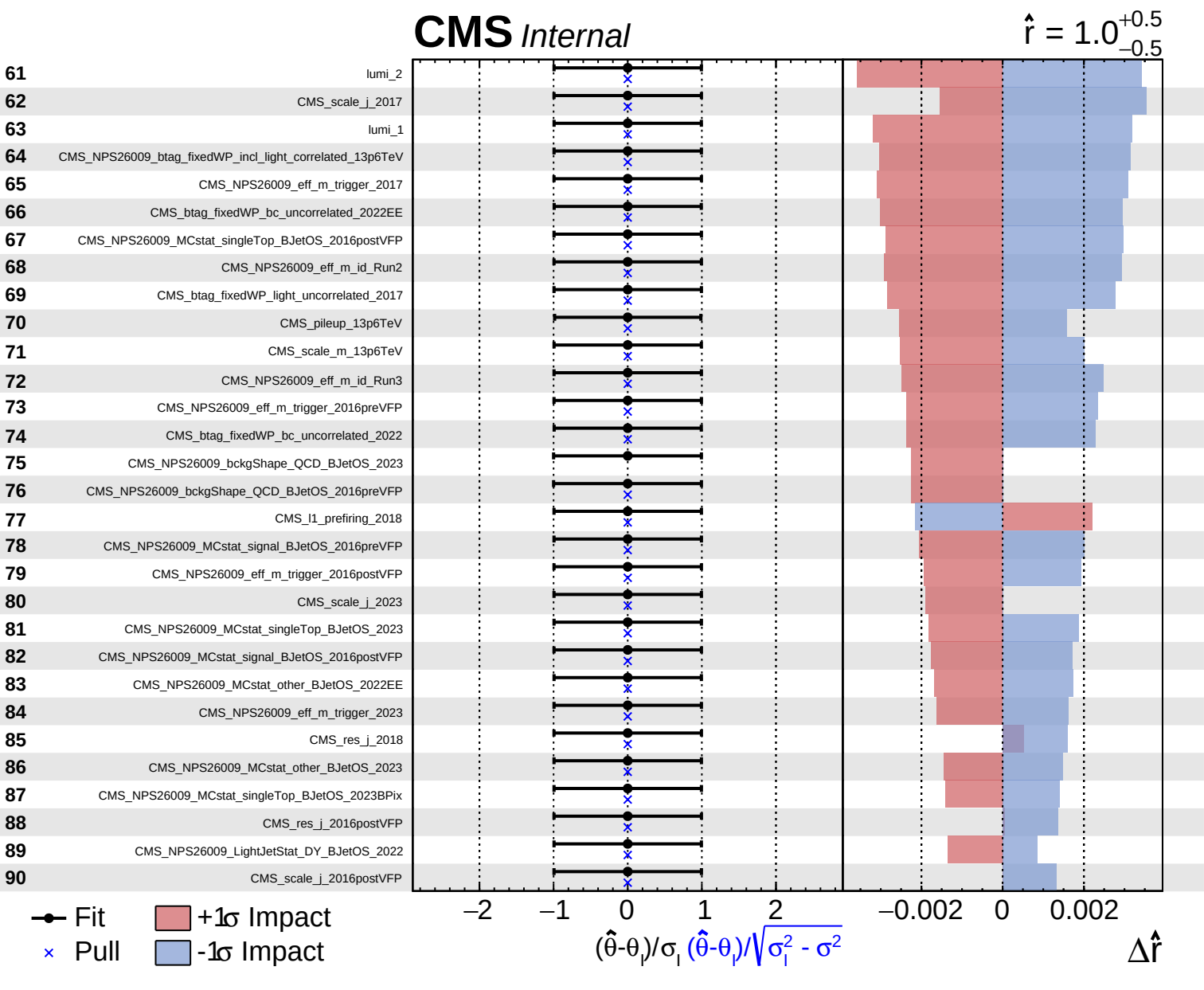
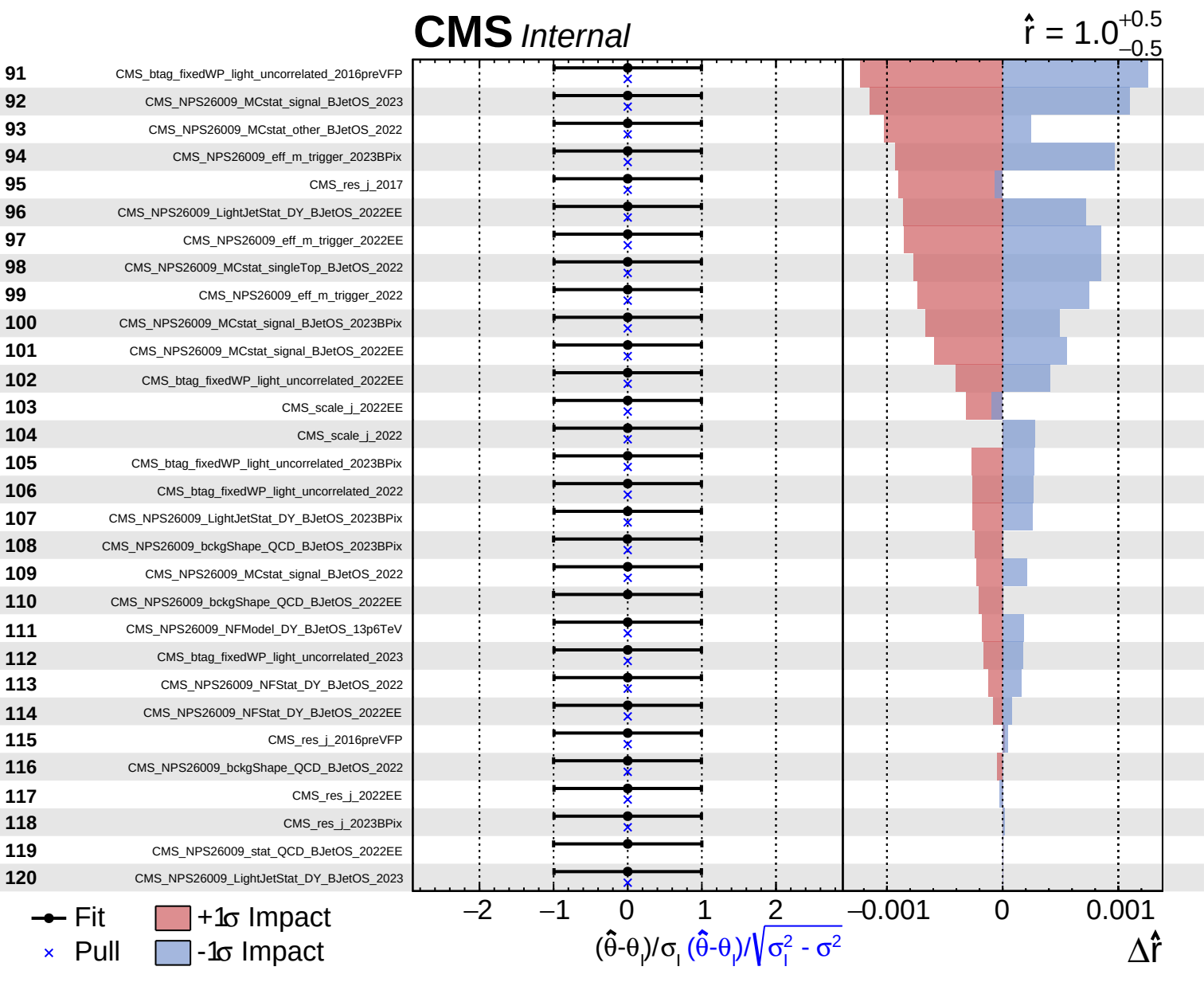
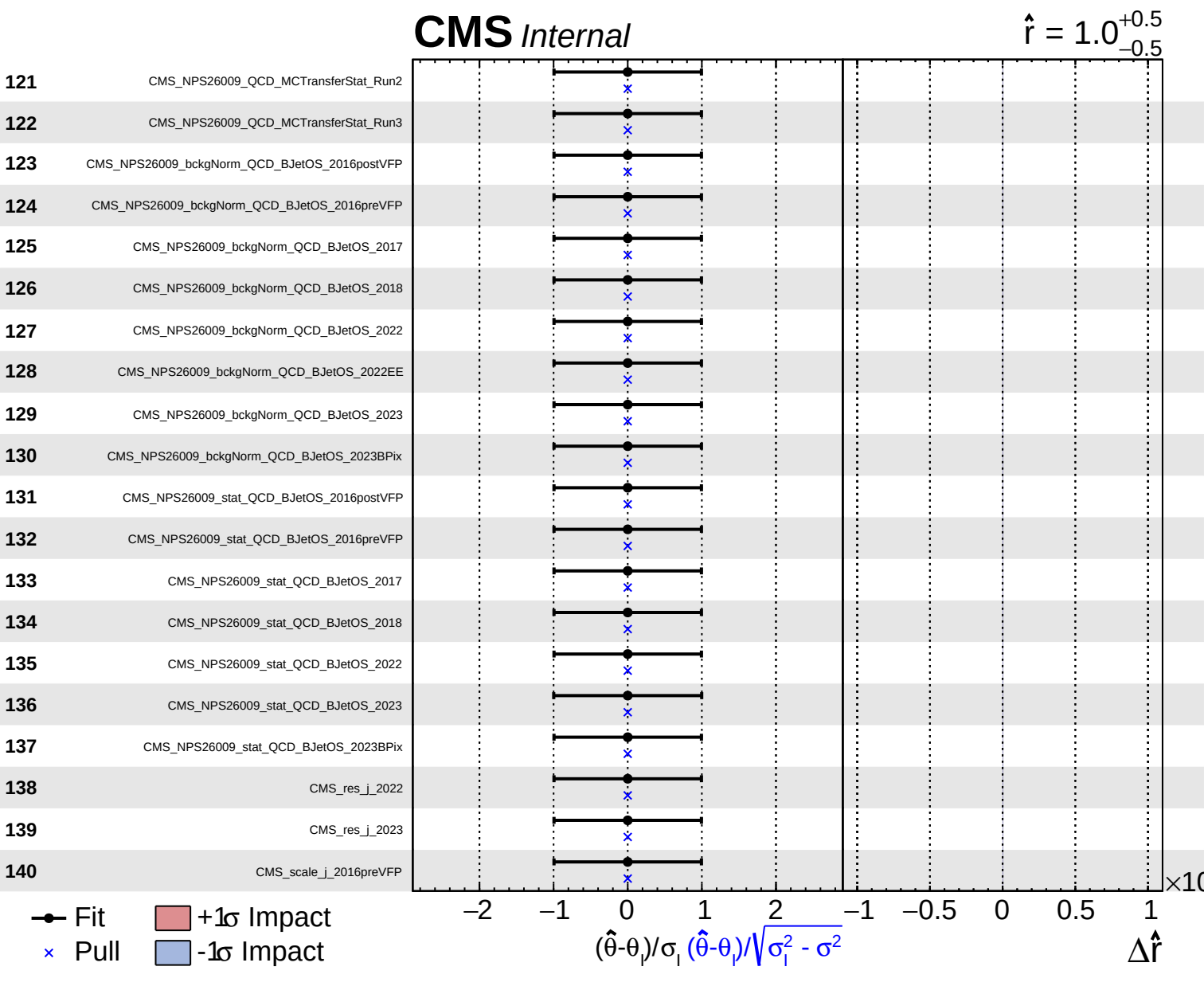

CMS Internal
$\hat{r} = 1.0^{+0.5}_{-0.5}$
1 QCDscale_ren_ttbar
2 QCDscale_fac_ttbar
3 CMS_NPS26009_NFModel_DY_BJetOS_13TeV
4 CMS_NPS26009_btag_fixedWP_comb_bc_correlated_13TeV
5 CMS_NPS26009_LightJetStat_DY_BJetOS_2018
6 CMS_NPS26009_pdf_NNPDF31
7 CMS_NPS26009_topmass_ttbar_BJetOS
8 pdf_alphas
9 CMS_NPS26009_NFStat_DY_BJetOS_2018
10 CMS_NPS26009_MCstat_ttbar_BJetOS_2018
11 CMS_NPS26009_LightJetStat_DY_BJetOS_2017
12 CMS_btag_fixedWP_bc_uncorrelated_2018
13 CMS_NPS26009_LightJetStat_DY_BJetOS_2016preVFP
14 CMS_NPS26009_btag_fixedWP_comb_bc_correlated_13p6TeV
15 CMS_NPS26009_MCstat_ttbar_BJetOS_2017
16 CMS_btag_fixedWP_bc_uncorrelated_2017
17 CMS_NPS26009_NFStat_DY_BJetOS_2017
18 CMS_NPS26009_MCstat_ttbar_BJetOS_2016preVFP
19 CMS_NPS26009_MCstat_ttbar_BJetOS_2016postVFP
20 CMS_NPS26009_LightJetStat_DY_BJetOS_2016postVFP
21 CMS_NPS26009_MCstat_ttbar_BJetOS_2023
22 lumi_13TeV_15161718_l
23 CMS_NPS26009_bckgShape_QCD_BJetOS_2017
24 CMS_NPS26009_NFStat_DY_BJetOS_2016preVFP
25 CMS_NPS26009_bckgShape_QCD_BJetOS_2018
26 CMS_NPS26009_btag_fixedWP_incl_light_correlated_13TeV
27 CMS_NPS26009_MCstat_ttbar_BJetOS_2023BPix
28 CMS_NPS26009_QCDscale_ren_singleTop
29 CMS_NPS26009_MCstat_ttbar_BJetOS_2022EE
30 CMS_NPS26009_MCstat_other_BJetOS_2018
Fit
Pull
+1σ Impact
-1σ Impact
−2 −1 0 1 2
$(\hat{\theta}-\theta_I)/\sigma_I$
$(\hat{\theta}-\theta_I)/\sqrt{\sigma_I^2 - \sigma^2}$
−0.2 −0.1 0 0.1 0.2
$\Delta\hat{r}$


CMS Internal
r̂ = 1.0 +0.5 −0.5
31 CMS_btag_fixedWP_bc_uncorrelated_2016preVFP
32 CMS_scale_m_13TeV
33 CMS_NPS26009_MCstat_singleTop_BJetOS_2018
34 CMS_pileup_13TeV
35 CMS_NPS26009_NFStat_DY_BJetOS_2016postVFP
36 CMS_btag_fixedWP_bc_uncorrelated_2016postVFP
37 CMS_NPS26009_MCstat_other_BJetOS_2016preVFP
38 CMS_NPS26009_MCstat_other_BJetOS_2016postVFP
39 CMS_NPS26009_MCstat_singleTop_BJetOS_2022EE
40 CMS_NPS26009_QCDscale_fac_singleTop
41 CMS_NPS26009_MCstat_singleTop_BJetOS_2017
42 CMS_NPS26009_MCstat_ttbar_BJetOS_2022
43 CMS_NPS26009_MCstat_other_BJetOS_2017
44 CMS_NPS26009_MCstat_other_BJetOS_2023BPix
45 lumi_13TeV_1516_l
46 CMS_NPS26009_MCstat_singleTop_BJetOS_2016preVFP
47 CMS_NPS26009_bckgShape_QCD_BJetOS_2016postVFP
48 CMS_l1_prefiring_2017
49 CMS_NPS26009_MCstat_signal_BJetOS_2018
50 CMS_btag_fixedWP_bc_uncorrelated_2023
51 CMS_scale_j_2023BPix
52 CMS_scale_j_2018
53 CMS_NPS26009_eff_m_trigger_2018
54 lumi_13TeV_151617_l
55 CMS_btag_fixedWP_light_uncorrelated_2018
56 CMS_NPS26009_MCstat_signal_BJetOS_2017
57 CMS_l1_prefiring_2016postVFP
58 CMS_btag_fixedWP_bc_uncorrelated_2023BPix
59 CMS_btag_fixedWP_light_uncorrelated_2016postVFP
60 CMS_l1_prefiring_2016preVFP
Fit
Pull
+1σ Impact
-1σ Impact
−2 −1 0 1 2
(θ̂-θ_I)/σ_I
−0.01 0 0.01
(θ̂-θ_I)/√(σ_I² - σ²)
Δr̂


CMS Internal
$\hat{r} = 1.0^{+0.5}_{-0.5}$
61 lumi_2
62 CMS_scale_j_2017
63 lumi_1
64 CMS_NPS26009_btag_fixedWP_incl_light_correlated_13p6TeV
65 CMS_NPS26009_eff_m_trigger_2017
66 CMS_btag_fixedWP_bc_uncorrelated_2022EE
67 CMS_NPS26009_MCstat_singleTop_BJetOS_2016postVFP
68 CMS_NPS26009_eff_m_id_Run2
69 CMS_btag_fixedWP_light_uncorrelated_2017
70 CMS_pileup_13p6TeV
71 CMS_scale_m_13p6TeV
72 CMS_NPS26009_eff_m_id_Run3
73 CMS_NPS26009_eff_m_trigger_2016preVFP
74 CMS_btag_fixedWP_bc_uncorrelated_2022
75 CMS_NPS26009_bckgShape_QCD_BJetOS_2023
76 CMS_NPS26009_bckgShape_QCD_BJetOS_2016preVFP
77 CMS_l1_prefiring_2018
78 CMS_NPS26009_MCstat_signal_BJetOS_2016preVFP
79 CMS_NPS26009_eff_m_trigger_2016postVFP
80 CMS_scale_j_2023
81 CMS_NPS26009_MCstat_singleTop_BJetOS_2023
82 CMS_NPS26009_MCstat_signal_BJetOS_2016postVFP
83 CMS_NPS26009_MCstat_other_BJetOS_2022EE
84 CMS_NPS26009_eff_m_trigger_2023
85 CMS_res_j_2018
86 CMS_NPS26009_MCstat_other_BJetOS_2023
87 CMS_NPS26009_MCstat_singleTop_BJetOS_2023BPix
88 CMS_res_j_2016postVFP
89 CMS_NPS26009_LightJetStat_DY_BJetOS_2022
90 CMS_scale_j_2016postVFP
Fit
Pull
+1σ Impact
-1σ Impact
−2
−1
0
1
2
$(\hat{\theta}-\theta_I)/\sigma_I$
−0.002
0
0.002
$(\hat{\theta}-\theta_I)/\sqrt{\sigma_I^2 - \sigma^2}$
$\Delta\hat{r}$


CMS Internal
r̂ = 1.0 +0.5 −0.5
91 CMS_btag_fixedWP_light_uncorrelated_2016preVFP
92 CMS_NPS26009_MCstat_signal_BJetOS_2023
93 CMS_NPS26009_MCstat_other_BJetOS_2022
94 CMS_NPS26009_eff_m_trigger_2023BPix
95 CMS_res_j_2017
96 CMS_NPS26009_LightJetStat_DY_BJetOS_2022EE
97 CMS_NPS26009_eff_m_trigger_2022EE
98 CMS_NPS26009_MCstat_singleTop_BJetOS_2022
99 CMS_NPS26009_eff_m_trigger_2022
100 CMS_NPS26009_MCstat_signal_BJetOS_2023BPix
101 CMS_NPS26009_MCstat_signal_BJetOS_2022EE
102 CMS_btag_fixedWP_light_uncorrelated_2022EE
103 CMS_scale_j_2022EE
104 CMS_scale_j_2022
105 CMS_btag_fixedWP_light_uncorrelated_2023BPix
106 CMS_btag_fixedWP_light_uncorrelated_2022
107 CMS_NPS26009_LightJetStat_DY_BJetOS_2023BPix
108 CMS_NPS26009_bckgShape_QCD_BJetOS_2023BPix
109 CMS_NPS26009_MCstat_signal_BJetOS_2022
110 CMS_NPS26009_bckgShape_QCD_BJetOS_2022EE
111 CMS_NPS26009_NFModel_DY_BJetOS_13p6TeV
112 CMS_btag_fixedWP_light_uncorrelated_2023
113 CMS_NPS26009_NFStat_DY_BJetOS_2022
114 CMS_NPS26009_NFStat_DY_BJetOS_2022EE
115 CMS_res_j_2016preVFP
116 CMS_NPS26009_bckgShape_QCD_BJetOS_2022
117 CMS_res_j_2022EE
118 CMS_res_j_2023BPix
119 CMS_NPS26009_stat_QCD_BJetOS_2022EE
120 CMS_NPS26009_LightJetStat_DY_BJetOS_2023
−2
−1
0
1
2
−0.001
0
0.001
Fit
Pull
+1σ Impact
-1σ Impact
(θ̂-θ_I)/σ_I
(θ̂-θ_I)/√(σ_I² - σ²)
Δr̂


CMS Internal
$\hat{r} = 1.0^{+0.5}_{-0.5}$
121 CMS_NPS26009_QCD_MCTransferStat_Run2
122 CMS_NPS26009_QCD_MCTransferStat_Run3
123 CMS_NPS26009_bckgNorm_QCD_BJetOS_2016postVFP
124 CMS_NPS26009_bckgNorm_QCD_BJetOS_2016preVFP
125 CMS_NPS26009_bckgNorm_QCD_BJetOS_2017
126 CMS_NPS26009_bckgNorm_QCD_BJetOS_2018
127 CMS_NPS26009_bckgNorm_QCD_BJetOS_2022
128 CMS_NPS26009_bckgNorm_QCD_BJetOS_2022EE
129 CMS_NPS26009_bckgNorm_QCD_BJetOS_2023
130 CMS_NPS26009_bckgNorm_QCD_BJetOS_2023BPix
131 CMS_NPS26009_stat_QCD_BJetOS_2016postVFP
132 CMS_NPS26009_stat_QCD_BJetOS_2016preVFP
133 CMS_NPS26009_stat_QCD_BJetOS_2017
134 CMS_NPS26009_stat_QCD_BJetOS_2018
135 CMS_NPS26009_stat_QCD_BJetOS_2022
136 CMS_NPS26009_stat_QCD_BJetOS_2023
137 CMS_NPS26009_stat_QCD_BJetOS_2023BPix
138 CMS_res_j_2022
139 CMS_res_j_2023
140 CMS_scale_j_2016preVFP
×10
Fit
Pull
+1σ Impact
-1σ Impact
−2 −1 0 1 2
$(\hat{\theta}-\theta_I)/\sigma_I$
−1 −0.5 0 0.5 1
$(\hat{\theta}-\theta_I)/\sqrt{\sigma_I^2 - \sigma^2}$
$\Delta\hat{r}$